

Year 7 · Chapter 4 · Fractions and percentages: answers (teacher copy)

4.01 Fractions (booklet p. 3)

Examples (model these live)

E1 Simplify $\frac{18}{24}$.

HCF of 18 and 24 is 6 $\therefore \frac{18}{24} = \frac{3}{4}$

E2 Write $2\frac{3}{4}$ as an improper fraction.

$2 \times 4 + 3 = 11 \therefore 2\frac{3}{4} = \frac{11}{4}$

E3 Write $\frac{17}{5}$ as a mixed numeral.

$17 \div 5 = 3 \text{ r } 2 \therefore \frac{17}{5} = 3\frac{2}{5}$

Questions (39 on today's skill, 17 from earlier lessons)

1 $\frac{1}{4}$

2 $\frac{3}{2}$

3 $1\frac{3}{8}$

4 $\frac{1}{5}$

5 $\frac{7}{5}$

6 $3\frac{1}{2}$

7 $\frac{1}{3}$

8 $\frac{18}{5}$

9 $b = 130$ 2.05

10 $\frac{7}{2}$

11 $2\frac{1}{2}$

12 $\frac{1}{6}$

13 39 3.03

14 $6 \times 4 = 24 \therefore \square = 5 \times 4 = 20$

15 $\frac{10}{3}$

16 $3\frac{2}{5}$

17 $1 \times 8, 2 \times 4 \therefore 1, 2, 4, 8$ 3.09

18 $\frac{3}{4}$

19 $12 \times 8 = 96 \therefore \square = 5 \times 8 = 40$

20 HCF of 32 and 40 is 8 $\therefore \frac{4}{5}$

21 3, 6, 9, 12, 15 3.12

22 $\frac{5}{10}$

23 $21 \div 5 = 4 \text{ r } 1 \therefore 4\frac{1}{5}$

24 divide, multiply, subtract, bring down $\therefore 212$ 3.05

25 $8 \times 4 = 32 \therefore \square = 5 \times 4 = 20$

26 HCF of 90 and 117 is 9 $\therefore \frac{10}{13}$

27 HCF of 8 and 64 is 8 $\therefore \frac{1}{8}$

28 $\frac{11}{12}$

29 acute 2.03

30 $19 \div 3 = 6 \text{ r } 1 \therefore 6\frac{1}{3}$

31 $\approx 5\,000 \div 80 \therefore \approx 63$ 3.01

32 $3 \times 9 = 27 \therefore \square = 2 \times 9 = 18$

33 $\times 6: \square = 24 \therefore \times 10: \square = 50$

34 $180^\circ - 142^\circ \therefore = 38^\circ$ 2.09

35 HCF of 8 and 48 is 8 $\therefore \frac{1}{6}$

36 90° 2.06

37 $\times 4: \square = 4 \therefore \times 10: \square = 50$

38 $\frac{12}{60} \therefore = \frac{1}{5}$

39 7; no 3.04

40 No. Halve the top and the bottom: $\frac{6}{10} = \frac{3}{5}$.

41 CC + LXXX + VIII $\therefore 200 + 80 + 8 = 288$ 3.06

42 $6 \times 3 + 1 \therefore = 19$

43 $\times 3: \square = 3 \therefore \times 10: \square = 40$

44 $2^2 \times 3$ 3.10

45 $\frac{40}{60} \therefore = \frac{2}{3}$

46 HCF of 2 $\therefore = 3/5$ 3.11

47 $\frac{15}{60} \therefore = \frac{1}{4}$

48 No. Halve the top and the bottom: $\frac{2}{8} = \frac{1}{4}$.

49 65° 2.08

50 $10 \times 4 + 3 \therefore = 43$

51 115° (check with a protractor) 2.02

52 $\times 4: \square = 4 \therefore \times 7: \square = 35$

53 $17 \div 8 = 2 \text{ r } 1 \therefore 2\frac{1}{8}$

54 154° 2.07

55 $\frac{36}{60} \therefore = \frac{3}{5}$

56 $= 8 + 4 \therefore = 12$ 3.08

4.02 Ordering fractions (booklet p. 5)

Examples (model these live)

E1 Which is larger: $\frac{3}{4}$ or $\frac{5}{8}$?

$\frac{3}{4} = \frac{6}{8} \therefore \frac{6}{8} > \frac{5}{8}$, so $\frac{3}{4}$

E2 Order from smallest to largest: $\frac{2}{3}$,

$\frac{1}{2}, \frac{5}{6}$
 $\frac{4}{6}, \frac{3}{6}, \frac{5}{6} \therefore \frac{1}{2}, \frac{2}{3}, \frac{5}{6}$

E3 Write < or >: $\frac{3}{5} \square \frac{2}{3}$

$\frac{9}{15}$ and $\frac{10}{15} \therefore \frac{3}{5} < \frac{2}{3}$

Questions (40 on today's skill, 17 from earlier lessons)

1 $\frac{3}{5}$

2 $\frac{1}{3}$

3 more

4 $\frac{6}{7}$

5 $\frac{1}{3}$

6 less

7 $\frac{3}{7}$

8 $\frac{1}{2}$

9 $\frac{35}{6}$ 4.01

10 $\frac{1}{8}$

11 more

12 $\frac{3}{4}$

13 6 3.05

14 $\frac{5}{60}$ and $\frac{54}{60} \therefore$ larger: $\frac{9}{10}$

15 $\frac{1}{6}$

16 less

17 $1\frac{1}{4}$ 4.01

18 $\frac{8}{9}$

19 $\frac{2}{6}$ and $\frac{1}{6} \therefore$ larger: $\frac{1}{3}$

20 $\frac{2}{6}$ and $\frac{3}{6} \therefore \frac{1}{3} < \frac{1}{2}$

21 $26 \div 3 = 8 \text{ r } 2 \therefore 8\frac{2}{3}$ 4.01

22 $\frac{5}{12}, \frac{8}{12}, \frac{4}{12} \therefore \frac{1}{3}, \frac{5}{12}, \frac{2}{3}$

23 $\frac{2}{4}$ and $\frac{3}{4} \therefore$ larger: $\frac{3}{4}$

24 they add to 180° 2.11

25 $\frac{5}{10}$ and $\frac{2}{10} \therefore \frac{1}{2} > \frac{1}{5}$

26 $\frac{4}{12}, \frac{7}{12}, \frac{3}{12} \therefore \frac{1}{4}, \frac{1}{3}, \frac{7}{12}$

27 $\frac{10}{12}$ and $\frac{9}{12} \therefore \frac{5}{6} > \frac{3}{4}$

28 $\frac{3}{12}, \frac{8}{12}, \frac{2}{12} \therefore \frac{1}{6}, \frac{1}{4}, \frac{2}{3}$

29 $\frac{49}{10}$ 4.01

30 $\frac{16}{24}$ and $\frac{21}{24} \therefore$ larger: $\frac{7}{8}$

31 HCF of 63 and 135 is 9 $\therefore = \frac{7}{15}$ 4.01

32 $\frac{20}{60}, \frac{108}{60}, \frac{100}{60}, \frac{80}{60} \therefore 1\frac{4}{5}, 1\frac{2}{3}, 1\frac{1}{3}, \frac{1}{3}$

33 $\frac{4}{6}$ and $\frac{5}{6} \therefore \frac{2}{3} < \frac{5}{6}$

34 list multiples of the largest; test the others $\therefore$ LCM = 60 3.12

35 $\frac{11}{12}, \frac{2}{12}, \frac{10}{12} \therefore \frac{1}{6}, \frac{5}{6}, \frac{11}{12}$

36 $\frac{3}{5}$ 4.01

37 $\frac{15}{20}$ and $\frac{16}{20} \therefore$ larger: $\frac{4}{5}$

38 $\frac{90}{60}, \frac{15}{60}, \frac{100}{60}, \frac{45}{60} \therefore 1\frac{2}{3}, 1\frac{1}{2}, \frac{3}{4}, \frac{1}{4}$

39 use a common denominator, then find one between $\therefore$ e.g. $\frac{19}{24}$

40 $8 \div 3 = 2 \text{ r } 2 \therefore 2\frac{2}{3}$ 4.01

41 $\frac{7}{8}$ is $\frac{1}{8}$ from 1; $\frac{5}{6}$ is $\frac{1}{6}$

from 1 $\therefore$ closer: $\frac{7}{8}$

42 $180^\circ - 57^\circ \therefore = 123^\circ$ 2.09

43 $\frac{40}{60}, \frac{100}{60}, \frac{90}{60}, \frac{42}{60} \therefore 1\frac{2}{3}, 1\frac{1}{2}, \frac{7}{10}, \frac{2}{3}$

44 use a common denominator, then find one between $\therefore$ e.g. $\frac{5}{12}$

45 $1\frac{3}{8}$ 4.01

46 $\frac{9}{10}$ is $\frac{1}{10}$ from 1; $\frac{11}{12}$ is $\frac{1}{12}$

from 1 $\therefore$ closer: $\frac{11}{12}$

47 HCF of 12 and 84 is 12 $\therefore = \frac{1}{7}$ 4.01

48 use a common denominator, then find one between $\therefore$ e.g. $\frac{7}{24}$

49 $\frac{9}{10}$ is $\frac{1}{10}$ from 1; $\frac{4}{5}$ is $\frac{1}{5}$

from 1 $\therefore$ closer: $\frac{9}{10}$

$$50 \quad 90^\circ \text{ 2.06}$$

$$51 \quad \frac{84}{60}, \frac{45}{60}, \frac{15}{60}, \frac{20}{60} \cdot 1\frac{2}{5}, \frac{3}{4}, \frac{1}{3}, \frac{1}{4}$$

$$52 \quad \frac{33}{8} \text{ 4.01}$$

$$53 \quad \frac{11}{12}, \frac{3}{12}, \frac{2}{12} \cdot \frac{1}{6}, \frac{1}{4}, \frac{11}{12}$$

54 use a common denominator, then find one between · e.g. $\frac{9}{20}$

$$55 \quad 32 \div 9 = 3 \text{ r } 5 \cdot = 3\frac{5}{9} \text{ 4.01}$$

$$56 \quad \frac{4}{5} \text{ is } \frac{1}{5} \text{ from } 1; \frac{11}{12} \text{ is } \frac{1}{12} \text{ from } 1 \cdot \text{ closer: } \frac{11}{12}$$

$$57 \quad 121^\circ \text{ 2.04}$$

4.03 Adding and subtracting fractions (booklet p. 7)

Examples (model these live)

$$E1 \quad \frac{3}{8} + \frac{1}{8} = \frac{4}{8} \cdot = \frac{1}{2}$$

$$E2 \quad \frac{2}{3} + \frac{1}{4} = \frac{8}{12} + \frac{3}{12} \cdot = \frac{11}{12}$$

$$E3 \quad \frac{5}{6} - \frac{1}{4} = \frac{10}{12} - \frac{3}{12} \cdot = \frac{7}{12}$$

Questions (46 on today's skill, 15 from earlier lessons)

$$1 \quad \frac{3}{8}$$

$$2 \quad \frac{1}{3}$$

$$3 \quad \frac{2}{3}$$

$$4 \quad \frac{10}{11}$$

$$5 \quad \frac{3}{11}$$

$$6 \quad \frac{1}{7}$$

$$7 \quad \frac{7}{8}$$

$$8 \quad \frac{2}{9}$$

$$9 \quad \frac{1}{9} \text{ 4.02}$$

$$10 \quad \frac{6}{11}$$

$$11 \quad \frac{8}{11}$$

$$12 \quad \frac{3}{5}$$

$$13 \quad \frac{1}{3}$$

$$14 \quad \frac{1}{3}$$

$$15 \quad \frac{2}{9}$$

$$16 \quad \frac{3}{11}$$

$$17 \quad \frac{5}{8}$$

$$18 \quad \frac{1}{3}$$

$$19 \quad 1$$

$$20 \quad \text{right 2.03}$$

$$21 \quad = \frac{12}{15} + \frac{14}{15} \cdot = 1\frac{11}{15}$$

$$22 \quad \frac{1}{10}$$

$$23 \quad \frac{2}{3}$$

$$24 \quad \frac{15}{4} \text{ 4.01}$$

$$25 \quad \frac{3}{4}$$

$$26 \quad = \frac{12}{15} - \frac{7}{15} \cdot = \frac{1}{3}$$

$$27 \quad \frac{1}{3}$$

$$28 \quad = \frac{3}{9} + \frac{8}{9} \cdot = 1\frac{2}{9}$$

$$29 \quad = \frac{10}{12} + \frac{9}{12} \cdot = 1\frac{7}{12}$$

$$30 \quad \frac{6}{12}, \frac{4}{12}, \frac{9}{12} \cdot \frac{1}{3}, \frac{1}{2}, \frac{3}{4} \text{ 4.02}$$

$$31 \quad \frac{1}{3} + \frac{1}{4} \cdot = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$$

$$32 \quad = \frac{9}{15} - \frac{7}{15} \cdot = \frac{2}{15}$$

$$33 \quad 17; \text{ no 3.04}$$

$$34 \quad = \frac{5}{20} + \frac{14}{20} \cdot = \frac{19}{20}$$

$$35 \quad = \frac{5}{10} + \frac{8}{10} \cdot = 1\frac{3}{10}$$

$$36 \quad \frac{1}{3} + \frac{1}{4} \cdot = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$$

$$37 \quad 3\frac{1}{2} \text{ 4.01}$$

$$38 \quad = \frac{9}{15} - \frac{1}{15} \cdot = \frac{8}{15}$$

$$39 \quad \frac{1}{10} \text{ and } \frac{2}{10} \cdot \text{ larger: } \frac{1}{5} \text{ 4.02}$$

$$40 \quad = \frac{3}{12} + \frac{4}{12} - \frac{1}{12} \cdot = \frac{6}{12} \cdot = \frac{1}{2}$$

$$41 \quad = \frac{5}{30} + \frac{3}{30} \cdot = \frac{4}{15}$$

$$42 \quad y = 180 - 90 = 90 \text{ (straight)} \cdot x = 90 \text{ (corresponding)} \text{ 2.10}$$

$$43 \quad \frac{1}{2} + \frac{1}{6} \cdot = \frac{3}{6} + \frac{1}{6} = \frac{2}{3}$$

$$44 \quad = \frac{4}{12} + \frac{3}{12} - \frac{2}{12} \cdot = \frac{5}{12} \cdot = \frac{5}{12}$$

$$45 \quad = \frac{4}{6} + \frac{5}{6} \cdot = \frac{9}{6} = 1\frac{1}{2}$$

$$46 \quad \frac{25}{40} \text{ and } \frac{36}{40} \cdot \frac{5}{8} < \frac{9}{10} \text{ 4.02}$$

$$47 \quad \frac{1}{5} - \frac{1}{6} \cdot = \frac{1}{30}$$

$$48 \quad 3 \text{ 3.05}$$

$$49 \quad \square = \frac{7}{8} - \frac{3}{4} \cdot \square = \frac{1}{8}$$

$$50 \quad = \frac{4}{12} + \frac{3}{12} - \frac{1}{12} \cdot = \frac{6}{12} \cdot = \frac{1}{2}$$

$$51 \quad \frac{13}{12} \text{ 4.01}$$

$$52 \quad = \frac{16}{24} + \frac{15}{24} \cdot = \frac{31}{24} = 1\frac{7}{24}$$

$$53 \quad \frac{3}{12}, \frac{11}{12}, \frac{1}{12} \cdot \frac{1}{12}, \frac{1}{4}, \frac{11}{12} \text{ 4.02}$$

$$54 \quad = \frac{16}{24} + \frac{9}{24} \cdot = \frac{25}{24} = 1\frac{1}{24}$$

$$55 \quad \frac{5}{6} - \frac{5}{8} \cdot = \frac{5}{24}$$

$$56 \quad a = 180 - 85 = 95 \cdot \text{ co-interior angles, } \parallel \text{ lines 2.09}$$

$$57 \quad \square = \frac{7}{8} - \frac{2}{3} \cdot \square = \frac{5}{24}$$

$$58 \quad 3\frac{1}{3} \text{ 4.01}$$

$$59 \quad = \frac{3}{12} + \frac{2}{12} - \frac{1}{12} \cdot = \frac{4}{12} \cdot = \frac{1}{3}$$

$$60 \quad \frac{1}{3} + \frac{1}{5} \cdot = \frac{5}{15} + \frac{3}{15} = \frac{8}{15}$$

$$61 \quad \frac{4}{12} \text{ and } \frac{5}{12} \cdot \text{ larger: } \frac{5}{12} \text{ 4.02}$$

4.04 Adding and subtracting mixed numerals (booklet p. 9)

Examples (model these live)

$$E1 \quad 2\frac{1}{5} + 1\frac{3}{5} = 3 + 1 = 4 \cdot \frac{1}{5} + \frac{3}{5} = \frac{4}{5} \cdot = 3\frac{4}{5}$$

$$E2 \quad 3\frac{1}{4} - 1\frac{3}{4} = \frac{13}{4} - \frac{7}{4} \cdot = \frac{6}{4} = 1\frac{1}{2}$$

$$E3 \quad 1\frac{2}{3} + 2\frac{1}{2} = \frac{5}{3} + \frac{5}{2} \cdot = \frac{10}{6} + \frac{15}{6} \cdot = \frac{25}{6} = 4\frac{1}{6}$$

Questions (47 on today's skill, 14 from earlier lessons)

$$1 \quad 3\frac{4}{7}$$

$$2 \quad 4\frac{1}{7}$$

$$3 \quad 4\frac{5}{7}$$

$$4 \quad 6\frac{3}{5}$$

$$5 \quad 2\frac{1}{9}$$

$$6 \quad 3\frac{5}{7}$$

$$7 \quad 8\frac{1}{2}$$

$$8 \quad 4\frac{1}{9}$$

$$9 \quad \frac{1}{3} \text{ 4.03}$$

$$10 \quad 4\frac{3}{4}$$

$$11 \quad 6\frac{7}{9}$$

$$12 \quad 3\frac{1}{5}$$

$$13 \quad 2\frac{3}{5}$$

$$14 \quad 5\frac{2}{3}$$

$$15 \quad 2\frac{2}{9}$$

$$16 \quad 5\frac{3}{5}$$

$$17 \quad 4\frac{1}{5}$$

$$18 \quad 4\frac{3}{7}$$

$$19 \quad 8\frac{2}{9}$$

$$20 \quad 360^\circ - 270^\circ \cdot = 90^\circ \text{ 2.02}$$

$$21 \quad = \frac{33}{8} - \frac{19}{8} \cdot = \frac{14}{8} = 1\frac{3}{4}$$

$$22 \quad 1\frac{2}{9}$$

$$23 \quad 2\frac{2}{3}$$

$$24 \quad \frac{1}{5} \text{ 4.02}$$

$$25 \quad 2\frac{3}{5}$$

$$26 \quad = \frac{29}{8} - \frac{23}{8} \cdot = \frac{6}{8} = \frac{3}{4}$$

$$27 \quad 2\frac{2}{5}$$

$$28 \quad = \frac{17}{5} - \frac{13}{5} \cdot = \frac{4}{5} = \frac{4}{5}$$

$$29 \quad 2 + 1 = 3 \cdot \frac{10}{15} + \frac{6}{15} = 1\frac{1}{15} \cdot = 4\frac{1}{15}$$

$$30 \quad \text{HCF of 24 and 56 is 8} \cdot = \frac{3}{7} \text{ 4.01}$$

$$31 \quad 6\frac{1}{2} - 2\frac{1}{2} \cdot = 4 \text{ m}$$

$$32 \quad = \frac{37}{6} - \frac{35}{6} \cdot = \frac{2}{6} = \frac{1}{3}$$

$$33 \quad \text{XXIII 3.06}$$

$$34 \quad 2 + 2 = 4 \cdot \frac{5}{10} + \frac{4}{10} = \frac{9}{10} \cdot = 4\frac{9}{10}$$

$$35 \quad 2 + 3 = 5 \cdot \frac{5}{10} + \frac{4}{10} = \frac{9}{10} \cdot = 5\frac{9}{10}$$

$$36 \quad 4\frac{1}{2} - 1\frac{1}{2} \cdot = 3 \text{ m}$$

$$37 \quad \frac{10}{11} \text{ 4.03}$$

$$38 \quad = \frac{53}{8} - \frac{39}{8} \cdot = \frac{14}{8} = 1\frac{3}{4}$$

$$39 \quad \frac{2}{12}, \frac{7}{12}, \frac{1}{12} \cdot \frac{1}{6}, \frac{1}{3}, \frac{7}{12} \text{ 4.02}$$

$$40 = \frac{15}{4} - \frac{5}{2} \cdot = \frac{15}{4} - \frac{10}{4} \cdot = 1\frac{1}{4}$$

$$41 = 5 + 4 = 9 \cdot \frac{4}{6} + \frac{5}{6} = 1\frac{1}{2} \cdot = 10\frac{1}{2}$$

$$42 = 12 - (-15) \cdot = 12 + 15 \cdot = 27^{\circ}\text{C} \text{ 1.10}$$

$$43 = 5\frac{1}{2} - 2\frac{1}{2} \cdot = 3 \text{ m}$$

$$44 = \frac{23}{4} - \frac{25}{6} \cdot = \frac{69}{12} - \frac{50}{12} \cdot = 1\frac{7}{12}$$

$$45 = 7\frac{1}{4} + 1\frac{3}{4} = 9 \cdot 2 = 18 \text{ m}$$

$$46 = \frac{6}{12} + \frac{5}{12} \cdot = \frac{11}{12} \text{ 4.03}$$

$$47 \text{ wholes: } 3 + 3 + 1 = 7 \cdot$$

$$\text{fractions: } \frac{7}{8} \cdot = 7\frac{7}{8}$$

$$48 \text{ 19; no } \text{ 3.04}$$

$$49 = \frac{13}{3} - \frac{5}{2} \cdot = \frac{26}{6} - \frac{15}{6} \cdot = 1\frac{5}{6}$$

$$50 = 7\frac{1}{2} + 1\frac{1}{2} = 9 \cdot 2 = 18 \text{ m}$$

$$51 = \frac{7}{12} \text{ 4.02}$$

$$52 \text{ wholes: } 3 + 1 + 2 = 6 \cdot$$

$$\text{fractions: } 1\frac{1}{12} \cdot = 7\frac{1}{12}$$

$$53 = 6\frac{1}{2} + 2\frac{1}{2} = 9 \cdot 2 = 18 \text{ m}$$

$$54 \text{ wholes: } 3 + 3 + 2 = 8 \cdot$$

$$\text{fractions: } \frac{7}{8} \cdot = 8\frac{7}{8}$$

$$55 = 3 + 4 \cdot = 7 \text{ 3.08}$$

$$56 = \frac{13}{3} - \frac{7}{4} \cdot = \frac{52}{12} - \frac{21}{12} \cdot = 2\frac{7}{12}$$

$$57 = \frac{1}{2} \text{ 4.03}$$

$$58 = 5\frac{1}{2} - 1\frac{1}{2} \cdot = 4 \text{ m}$$

$$59 = 6\frac{3}{4} + 4\frac{3}{4} = 11\frac{1}{2} \cdot \times 2 = 23 \text{ m}$$

$$60 = \frac{7}{10} \text{ and } \frac{5}{10} \cdot \frac{7}{10} > \frac{1}{2} \text{ 4.02}$$

$$61 \text{ wholes: } 2 + 1 + 2 = 5 \cdot$$

$$\text{fractions: } \frac{11}{12} \cdot = 5\frac{11}{12}$$

4.05 Fraction of a quantity (booklet p. 11)

Examples (model these live)

E1 $\frac{1}{5}$ of 35
 $35 \div 5 \cdot = 7$

E2 $\frac{3}{4}$ of 20
 $20 \div 4 = 5 \cdot 5 \times 3 = 15$

E3 $\frac{2}{3}$ of an hour, in minutes
 $1 \text{ hour} = 60 \text{ minutes} \cdot 60 \div 3 \times 2 = 40 \text{ minutes}$

Questions (44 on today's skill, 16 from earlier lessons)

1 5
 2 4
 3 3
 4 11
 5 27
 6 15
 7 8
 8 15
 9 $\frac{7}{4}$ 4.01
 10 2
 11 10
 12 15
 13 11
 14 16
 15 14
 16 12
 17 parallel 2.06
 18 $18 \div 3 = 6 \cdot 6 \times 2 = 12$
 19 10
 20 10

21 $1\frac{1}{5}$ 4.04
 22 4
 23 $48 \div 8 = 6 \cdot 6 \times 7 = 42$
 24 $1 \text{ kg} = 1000 \text{ grams} \cdot 1000 \div 10 \times 3 = 300 \text{ grams}$
 25 $= \frac{10}{15} - \frac{6}{15} \cdot = \frac{4}{15}$ 4.03
 26 $24 \div 4 \times 3 \cdot = 18 \text{ students}$
 27 $30 \div 3 = 10 \cdot 10 \times 2 = 20$
 28 $1 \times 15, 3 \times 5 \cdot 1, 3, 5, 15$ 3.09
 29 $\text{a day} = 24 \text{ hours} \cdot 24 \div 3 \times 2 = 16 \text{ hours}$
 30 $25 \div 5 \times 3 \cdot = 15 \text{ players}$
 31 $\$1 = 100 \text{ cents} \cdot 100 \div 2 \times 1 = 50 \text{ cents}$
 32 $30 \div 5 \times 4 \cdot = 24 \text{ eggs}$
 33 $\frac{1}{3}$ 4.02
 34 $240 \div 10 = 24 \cdot 24 \times 9 = 216$
 35 $\frac{7}{8}$ 4.01

36 $15 \times 7 \cdot = 105$
 37 $\text{an hour} = 60 \text{ minutes} \cdot 60 \div 2 \times 1 = 30 \text{ minutes}$
 38 $\text{short division, left to right} \cdot = 249$ 3.03
 39 $24 \div 6 \times 1 \cdot = 4 \text{ players}$
 40 $1\frac{1}{6}$ 4.04
 41 $57 \div 3 = 19 \cdot 19 \times 2 = 38$
 42 $6 \times 3 \cdot = 18$
 43 $\frac{1}{5}$ of it is $9 \div 3 = 3 \cdot$ the number: $3 \times 5 = 15$
 44 $\frac{1}{3} + \frac{1}{5} \cdot = \frac{5}{15} + \frac{3}{15} = \frac{8}{15}$ 4.03
 45 $\frac{2}{3}$ of 60 = 40, $\frac{3}{4}$ of 36 = 27 $\cdot$ more: $\frac{2}{3}$ of 60
 46 acute 2.03
 47 $\text{discount: } 12 \div 3 = \$4 \cdot$ price: $12 - 4 = \$8$

48 $\text{food } \$20, \text{ transport } \$15 \cdot$ left: $60 - 20 - 15 = \$25$
 49 less 4.02
 50 $14 \times 4 \cdot = 56$
 51 $\frac{1}{3}$ of it is $12 \div 2 = 6 \cdot$ the number: $6 \times 3 = 18$
 52 $\frac{2}{3}$ of 36 = 24, $\frac{3}{4}$ of 72 = 54 $\cdot$ more: $\frac{3}{4}$ of 72
 53 $\text{CD} + \text{XXX} \cdot 400 + 30 = 430$ 3.06
 54 $\text{discount: } 72 \div 3 = \$24 \cdot$ price: $72 - 24 = \$48$
 55 $5\frac{4}{5}$ 4.04
 56 $\text{food } \$8, \text{ transport } \$6 \cdot$ left: $24 - 8 - 6 = \$10$
 57 $10 \times 5 \cdot = 50$
 58 $= \frac{9}{15} + \frac{1}{15} \cdot = \frac{2}{3}$ 4.03
 59 $20 \div 5 \times 1 \cdot = 4 \text{ students}$
 60 $180^{\circ} - 106^{\circ} \cdot = 74^{\circ}$ 2.09

4.06 Multiplying fractions (booklet p. 13)

Examples (model these live)

E1 $\frac{2}{3} \times \frac{4}{5}$
 $2 \times 4 = 8, 3 \times 5 = 15 \cdot = \frac{8}{15}$

E2 $\frac{3}{4} \times \frac{8}{9}$
 cancel: $\frac{1}{1} \times \frac{2}{3} \cdot = \frac{2}{3}$

E3 $1\frac{1}{2} \times 2\frac{2}{3}$
 $= \frac{3}{2} \times \frac{8}{3} \cdot = 4$

Questions (49 on today's skill, 17 from earlier lessons)

1 $\frac{1}{45}$
 2 $3\frac{6}{7}$
 3 $\frac{1}{16}$
 4 $\frac{8}{35}$
 5 4
 6 $\frac{1}{8}$
 7 $\frac{7}{12}$
 8 1
 9 15 4.05
 10 $\frac{1}{15}$

11 $\frac{2}{7}$
 12 4
 13 $\frac{1}{12}$
 14 $\frac{2}{81}$
 15 $1\frac{1}{5}$
 16 $\frac{1}{12}$
 17 3
 18 $\frac{1}{4}$
 19 $\frac{2}{9}$
 20 4 3.07

21 cancel 3: $\frac{1}{7} \times \frac{4}{7} \cdot = \frac{4}{49}$
 22 $1\frac{4}{5}$
 23 $\frac{1}{18}$
 24 $3\frac{3}{5}$ 4.04
 25 $\frac{1}{16}$
 26 cancel 3: $\frac{1}{7} \times \frac{1}{3} \cdot = \frac{1}{21}$
 27 6
 28 cancel 2: $\frac{3}{7} \times \frac{3}{5} \cdot = \frac{9}{35}$
 29 $= \frac{10}{3} \times \frac{3}{5} \cdot = 2$

30 HCF of 16 and 24 is $8 \cdot = \frac{2}{3}$ 4.01
 31 $= 16 \div 4 \times 1 \cdot = 4$
 32 cancel 5: $\frac{1}{7} \times \frac{1}{6} \cdot = \frac{1}{42}$
 33 $= \frac{12}{15} + \frac{10}{15} \cdot = 1\frac{7}{15}$ 4.03
 34 $= \frac{17}{5} \times \frac{1}{3} \cdot = 1\frac{2}{15}$
 35 $= 12 \div 6 \times 1 \cdot = 2$
 36 $= \frac{5}{2} \times \frac{1}{2} \cdot = 1\frac{1}{4}$
 37 $= 35 \div 7 \times 1 \cdot = 5$
 38 103° 2.08

$$39 \text{ cancel 2: } \frac{1}{5} \times \frac{1}{4} \cdot = \frac{1}{20}$$

$$40 \frac{1}{7} \text{ 4.02}$$

$$41 = \frac{3}{2} \times \frac{5}{3} \cdot = \frac{15}{6} \cdot = 2\frac{1}{2}$$

$$42 = \frac{13}{4} \times \frac{2}{5} \cdot = 1\frac{3}{10}$$

$$43 \text{ 14 4.05}$$

$$44 = 6 \div 3 \times 1 \cdot = 2$$

$$45 5\frac{1}{2} - 2\frac{1}{2} \cdot = 3 \text{ m 4.04}$$

$$46 \text{ cancel 4: } \frac{1}{7} \times \frac{1}{6} \cdot = \frac{1}{42}$$

$$47 = \frac{10}{3} \times \frac{17}{5} \cdot = \frac{170}{15} \cdot = 11\frac{1}{3}$$

$$\frac{1}{3}$$

$$48 \frac{7}{2} \times \frac{3}{2} \cdot = 5\frac{1}{4} \text{ m}^2$$

$$49 = 8 + 6 \cdot = 14 \text{ 3.08}$$

50 Smaller: multiplying by a fraction less than 1 gives a smaller answer ($\frac{3}{10}$).

$$51 \frac{3}{4} \text{ 4.03}$$

$$52 = \frac{18}{5} \times \frac{16}{5} \cdot = \frac{288}{25} \cdot = 11\frac{13}{25}$$

$$53 \frac{9}{2} \times \frac{7}{2} \cdot = 15\frac{3}{4} \text{ m}^2$$

$$54 \frac{5}{6} \text{ 4.01}$$

55 Smaller: multiplying by a fraction less than 1 gives a smaller answer ($\frac{2}{3}$).

$$56 100 \div 5 = 20 \cdot 20 \times 2 = 40 \text{ 4.05}$$

$$57 \frac{16}{3} \times \frac{5}{3} \cdot = 8\frac{8}{9} \text{ m}^2$$

58 Smaller: multiplying by a fraction less than 1 gives a smaller answer ($\frac{1}{2}$).

$$59 99^\circ \text{ 2.04}$$

$$60 = \frac{3}{2} \times \frac{7}{4} \cdot = \frac{21}{8} \cdot = 2\frac{5}{8}$$

$$61 \text{ more 4.02}$$

$$62 = 10 \div 5 \times 3 \cdot = 6$$

$$63 \frac{13}{3} \times \frac{11}{3} \cdot = 15\frac{8}{9} \text{ m}^2$$

$$64 4\frac{3}{7} \text{ 4.04}$$

$$65 = \frac{11}{3} \times \frac{9}{4} \cdot = \frac{99}{12} \cdot = 8\frac{1}{4}$$

$$66 = \frac{12}{15} + \frac{13}{15} \cdot = 1\frac{2}{3} \text{ 4.03}$$

4.07 Dividing fractions (booklet p. 15)

Examples (model these live)

E1 Write the reciprocal of $\frac{3}{7}$.

turn it upside down $\cdot = \frac{7}{3}$

$$\text{E2 } \frac{2}{3} \div \frac{4}{5}$$

$$= \frac{2}{3} \times \frac{5}{4} \cdot = \frac{10}{12} = \frac{5}{6}$$

$$\text{E3 } 2\frac{1}{2} \div 1\frac{1}{4}$$

$$= \frac{5}{2} \div \frac{5}{4} \cdot = \frac{5}{2} \times \frac{4}{5} \cdot = 2$$

Questions (38 on today's skill, 16 from earlier lessons)

$$1 \frac{6}{11}$$

$$2 \frac{30}{1}$$

$$3 \frac{1}{35}$$

$$4 \frac{4}{3}$$

$$5 \frac{8}{21}$$

$$6 \frac{4}{21}$$

$$7 \frac{9}{2}$$

$$8 \frac{10}{4}$$

$$9 \frac{3}{4} \text{ 4.06}$$

$$10 \frac{12}{32}$$

$$11 \frac{7}{32}$$

$$12 \frac{7}{6}$$

$$13 141^\circ \text{ 2.04}$$

$$14 = \frac{3}{7} \times \frac{4}{1} \cdot = 1\frac{5}{7}$$

$$15 \frac{12}{27}$$

$$16 \frac{8}{27}$$

$$17 \text{ 20 4.05}$$

$$18 \frac{12}{5}$$

$$19 = \frac{1}{7} \times \frac{7}{6} \cdot = \frac{1}{6}$$

$$20 4 \div \frac{1}{2} = 4 \times 2 \cdot = 8 \text{ scoops}$$

$$21 \text{ HCF of 8 and 44 is } 4 \cdot = \frac{2}{11} \text{ 4.01}$$

$$22 = 5 \times \frac{3}{2} \cdot = 7\frac{1}{2}$$

$$23 = \frac{1}{8} \times \frac{8}{1} \cdot = 1$$

$$24 3 + 3 = 6 \cdot \frac{9}{12} + \frac{10}{12} = 1\frac{7}{12} \cdot = 7\frac{7}{12} \text{ 4.04}$$

$$25 2 \div \frac{1}{3} = 2 \times 3 \cdot = 6 \text{ scoops}$$

$$26 3 \div \frac{1}{2} = 3 \times 2 \cdot = 6 \text{ scoops}$$

$$27 = 6 \times \frac{5}{4} \cdot = 7\frac{1}{2}$$

$$28 b = 85 \text{ 2.05}$$

$$29 = \frac{1}{3} \times \frac{7}{4} \cdot = \frac{7}{12}$$

$$30 \frac{1}{6} \text{ 4.02}$$

$$31 = \frac{13}{4} \div \frac{5}{3} \cdot = \frac{13}{4} \times \frac{3}{5} \cdot = 1\frac{19}{20}$$

$$32 5 \div \frac{1}{2} = 5 \times 2 \cdot = 10 \text{ scoops}$$

$$33 \frac{1}{20} \text{ 4.06}$$

$$34 = 5 \times \frac{6}{5} \cdot = 6$$

$$35 35 \div 5 \times 2 \cdot = 14 \text{ eggs 4.05}$$

$$36 = \frac{5}{4} \div \frac{11}{4} \cdot = \frac{5}{4} \times \frac{4}{11} \cdot = \frac{5}{11}$$

$$37 6 \div \frac{3}{4} \cdot = 6 \times \frac{4}{3} = 8 \text{ pieces}$$

$$38 = 5 \times 5 \times 5 \times 5 \cdot = 625 \text{ 3.07}$$

39 Larger: dividing by a fraction less than 1 gives a larger answer (2).

$$40 2\frac{1}{2} \text{ 4.04}$$

$$41 = \frac{5}{2} \div \frac{8}{3} \cdot = \frac{5}{2} \times \frac{3}{8} \cdot = \frac{15}{16}$$

$$42 7\frac{1}{2} \div \frac{3}{4} \cdot = \frac{15}{2} \times \frac{4}{3} = 10$$

$$\text{pieces}$$

$$43 = \frac{3}{6} - \frac{2}{6} \cdot = \frac{1}{6} \text{ 4.03}$$

44 Larger: dividing by a fraction less than 1 gives a larger answer ($1\frac{1}{2}$).

$$45 4\frac{1}{2} \div \frac{3}{8} \cdot = \frac{9}{2} \times \frac{8}{3} = 12$$

$$\text{pieces}$$

46 Larger: dividing by a fraction less than 1 gives a larger answer ($1\frac{2}{3}$).

$$47 91 \text{ 000 3.02}$$

$$48 = \frac{5}{2} \div \frac{8}{5} \cdot = \frac{5}{2} \times \frac{5}{8} \cdot = 1\frac{9}{16}$$

$$49 3\frac{3}{5} \text{ 4.01}$$

$$50 = 2 \times \frac{7}{6} \cdot = 2\frac{1}{3}$$

$$51 10 \div \frac{2}{3} \cdot = 10 \times \frac{3}{2} = 15$$

$$\text{pieces}$$

$$52 3 \text{ 4.05}$$

$$53 = \frac{12}{5} \div \frac{14}{5} \cdot = \frac{12}{5} \times \frac{5}{14} \cdot = \frac{6}{7}$$

$$54 = \frac{21}{5} - \frac{14}{5} \cdot = \frac{7}{5} = 1\frac{2}{5}$$

$$\text{4.04}$$

4.08 Fraction problems (booklet p. 17)

Examples (model these live)

E1 Sam had \$60 and spent $\frac{2}{5}$ of it.

How much is left?

spent: $60 \div 5 \times 2 = \$24$ $\cdot$ left: $60 - 24 = \$36$

E2 In a class, $\frac{1}{2}$ walk and $\frac{1}{3}$ ride. The

rest take the bus. What fraction take the bus?

$$\frac{1}{2} + \frac{1}{3} = \frac{5}{6} \cdot \text{ bus: } 1 - \frac{5}{6} = \frac{1}{6}$$

E3 How many $\frac{3}{4}$ m pieces can be cut

from 6 m?

$$6 \div \frac{3}{4} = 6 \times \frac{4}{3} \cdot = 8 \text{ pieces}$$

Questions (34 on today's skill, 12 from earlier lessons)

$$1 \frac{1}{5}$$

$$2 \frac{10}{20} \cdot = \frac{1}{2}$$

$$3 \frac{1}{3}, 7 \text{ cm}$$

$$4 \frac{2}{3}$$

$$5 \frac{8}{17} \cdot = \frac{8}{17}$$

$$6 \frac{1}{2}, 9 \text{ cm}$$

$$7 \frac{1}{5}$$

$$8 \frac{9}{23} \cdot = \frac{9}{23}$$

$$9 \frac{1}{3}, 9 \text{ cm}$$

$$10 \frac{1}{3}$$

$$11 \text{ they are equal 2.11}$$

$$12 \text{ spent: } 36 \div 3 \times 1 = \$12 \cdot \text{ left: } 36 - 12 = \$24$$

$$13 \frac{15}{29} \cdot = \frac{15}{29}$$

$$14 \frac{1}{3}, 3 \text{ cm}$$

$$15 \text{ spent: } 20 \div 4 \times 1 = \$5 \cdot \text{ left: } 20 - 5 = \$15$$

$$16 \text{ walk 20, ride 8} \cdot \text{ bus: } 40 - 20 - 8 = 12$$

$$17 \text{ HCF of 66 and 72 is } 6 \cdot = \frac{11}{12} \text{ 4.01}$$

$$18 \frac{1}{4} + \frac{1}{6} = \frac{5}{12} \cdot \text{ flowers: } 1 - \frac{5}{12} = \frac{7}{12}$$

$$\frac{5}{12} = \frac{7}{12}$$

$$19 \text{ spent: } 20 \div 4 \times 3 = \$15 \cdot \text{ left: } 20 - 15 = \$5$$

$$20 = \frac{9}{5} \times \frac{1}{2} \cdot = \frac{9}{10} \text{ 4.06}$$

$$21 \text{ walk 20, ride 16} \cdot \text{ bus: } 80 - 20 - 16 = 44$$

$$22 \quad \frac{1}{2} + \frac{1}{8} = \frac{5}{8} \cdot \text{flowers: } 1 - \frac{5}{8} = \frac{3}{8}$$

$$23 \quad 172^\circ \text{ 2.04}$$

$$24 \quad \text{spent: } 33 \div 3 \times 1 = \$11 \cdot \text{left: } 33 - 11 = \$22$$

$$25 \quad \frac{1}{3} \text{ 4.02}$$

$$26 \quad \text{Mon: } 48 \div 3 = 16, \text{ left } 32 \cdot \text{Tue: } 32 \div 4 = 8 \cdot \text{left: } \$24$$

$$27 \quad \text{walk 25, ride 20} \cdot \text{bus: } 100 - 25 - 20 = 55$$

$$28 \quad 2 \text{ 4.05}$$

$$29 \quad \frac{1}{2} + \frac{1}{3} = \frac{5}{6} \cdot \text{flowers: } 1 - \frac{5}{6} = \frac{1}{6}$$

$$30 \quad \text{Mon: } 120 \div 3 = 40, \text{ left } 80 \cdot \text{Tue: } 80 \div 4 = 20 \cdot \text{left: } \$60$$

$$31 \quad \text{left} = \frac{2}{5} \text{ of the money} \cdot \frac{1}{5} = 120 \div 2 = 60 \cdot \text{start: } 60 \times 5 = \$300$$

$$32 \quad = 5 \times 5 \times 5 \times 5 = 625 \text{ 3.07}$$

$$33 \quad 9 \div \frac{3}{4} = 9 \times \frac{4}{3} = 12 \text{ lengths}$$

$$34 \quad \frac{1}{30} \text{ 4.06}$$

$$35 \quad \text{Mon: } 84 \div 3 = 28, \text{ left } 56 \cdot \text{Tue: } 56 \div 4 = 14 \cdot \text{left: } \$42$$

$$36 \quad \text{left} = \frac{2}{3} \text{ of the money} \cdot \frac{1}{3} = 200 \div 2 = 100 \cdot \text{start: } 100 \times 3 = \$300$$

$$37 \quad = \frac{12}{20} + \frac{15}{20} = 1 \frac{7}{20} \text{ 4.03}$$

$$38 \quad \text{left} = \frac{3}{4} \text{ of the money} \cdot \frac{1}{4} = 60 \div 3 = 20 \cdot \text{start: } 20 \times 4 = \$80$$

$$39 \quad 12 \div \frac{2}{3} = 12 \times \frac{3}{2} = 18 \text{ lengths}$$

$$40 \quad 2 \text{ 3.08}$$

$$41 \quad \text{Mon: } 120 \div 3 = 40, \text{ left } 80 \cdot \text{Tue: } 80 \div 4 = 20 \cdot \text{left: } \$60$$

$$42 \quad 4 \frac{1}{3} \text{ 4.04}$$

$$43 \quad \frac{1}{2} + \frac{1}{6} = \frac{2}{3} \cdot \text{flowers: } 1 - \frac{2}{3} = \frac{1}{3}$$

$$44 \quad \text{left} = \frac{1}{4} \text{ of the money} \cdot \frac{1}{4} = 20 \div 1 = 20 \cdot \text{start: } 20 \times 4 = \$80$$

$$45 \quad \frac{4}{7} \text{ 4.07}$$

$$46 \quad 15 \div \frac{3}{5} = 15 \times \frac{5}{3} = 25 \text{ lengths}$$

4.09 Fractions on a calculator (booklet p. 19)

Examples (model these live)

E1 Use a calculator: $\frac{5}{12} + \frac{7}{18}$

$$5 \square / \square 12 + 7 \square / \square 18 = \cdot = \frac{29}{36}$$

E2 Use a calculator: $2\frac{3}{8} \times 1\frac{5}{6}$

$$\text{enter each mixed numeral } \cdot = 4 \frac{17}{48}$$

E3 Use a calculator to write $\frac{7}{16}$ as a decimal.

$$7 \square / \square 16 = \text{then } S \Rightarrow D \cdot = 0.4375$$

Questions (40 on today's skill, 16 from earlier lessons)

$$1 \quad \frac{8}{117}$$

$$2 \quad 0.375$$

$$3 \quad \frac{11}{13}$$

$$4 \quad 1 \frac{45}{56}$$

$$5 \quad 0.1875$$

$$6 \quad \frac{1}{9}$$

$$7 \quad \frac{3}{5}$$

$$8 \quad 0.36$$

$$9 \quad 3 \text{ 4.06}$$

$$10 \quad 0.875$$

$$11 \quad \frac{7}{15}$$

$$12 \quad \frac{65}{196}$$

$$13 \quad 7 \text{ 000 3.01}$$

$$14 \quad 15 \square / \square 16 \div 4 \square / \square 7 = \cdot =$$

$$1 \frac{41}{64}$$

$$15 \quad 0.26$$

$$16 \quad \frac{10}{11}$$

$$17 \quad 10 \text{ 4.07}$$

$$18 \quad 1 \frac{71}{95}$$

$$19 \quad 4 \square / \square 17 \div 7 \square / \square 9 = \cdot =$$

$$\frac{36}{119}$$

$$20 \quad \text{enter each mixed numeral} \cdot = 2 \frac{2}{5}$$

$$21 \quad \text{HCF of 72 and 117 is } 9 \cdot = \frac{8}{13} \text{ 4.01}$$

$$22 \quad \text{bracket: } 1 \frac{1}{6} \cdot = \frac{35}{48}$$

$$23 \quad 1 \square / \square 7 \div 9 \square / \square 11 = \cdot = \frac{11}{63}$$

$$24 \quad \text{walk 15, ride 9} \cdot \text{bus: } 45 - 15 - 9 = 21 \text{ 4.08}$$

$$25 \quad \text{enter each mixed numeral} \cdot = 3 \frac{1}{9}$$

$$26 \quad \text{bracket: } \frac{23}{24} \cdot = \frac{23}{32}$$

$$27 \quad \text{enter each mixed numeral} \cdot = 1 \frac{1}{4}$$

$$28 \quad \text{bracket: } 1 \frac{1}{8} \cdot = 1$$

$$29 \quad 180^\circ - 126^\circ \cdot = 54^\circ \text{ 2.09}$$

$$30 \quad 1 \square / \square 5 \div 6 \square / \square 19 = \cdot =$$

$$\frac{19}{30}$$

$$31 \quad \frac{1}{4} \text{ 4.02}$$

$$32 \quad 0.6923 \text{ and } 0.6875 \cdot \text{larger: } \frac{9}{13}$$

$$33 \quad \text{enter each mixed numeral} \cdot = 1 \frac{1}{3}$$

$$34 \quad \frac{1}{30} \text{ 4.06}$$

$$35 \quad \text{bracket: } \frac{19}{42} \cdot = \frac{19}{168}$$

$$36 \quad = 6 \times \frac{5}{4} = 7 \frac{1}{2} \text{ 4.07}$$

$$37 \quad 1 \square / \square 6 \div 3 \square / \square 16 = \cdot = \frac{8}{9}$$

$$38 \quad 0.7778 \text{ and } 0.7857 \cdot \text{larger: } \frac{11}{14}$$

$$39 \quad 2 \frac{1}{3} \times 12 \cdot = 28 \text{ kg}$$

$$40 \quad 115^\circ \text{ (check with a protractor) 2.02}$$

$$41 \quad \text{enter each fraction} \cdot = \frac{1}{4}$$

$$42 \quad \frac{1}{2}, 7 \text{ cm 4.08}$$

$$43 \quad 0.7143 \text{ and } 0.7273 \cdot \text{larger: } \frac{8}{11}$$

$$44 \quad 3 \frac{1}{3} \times 29 \cdot = 96 \frac{2}{3} \text{ kg}$$

$$45 \quad = \frac{15}{20} - \frac{8}{20} \cdot = \frac{7}{20} \text{ 4.03}$$

$$46 \quad \text{enter each fraction} \cdot = \frac{19}{72}$$

$$47 \quad 5 \frac{1}{8} \times 12 \cdot = 61 \frac{1}{2} \text{ kg}$$

$$48 \quad \text{enter each fraction} \cdot = \frac{2}{7}$$

$$49 \quad \angle WBD \text{ or } \angle DBW \text{ 2.01}$$

$$50 \quad 0.7647 \text{ and } 0.7619 \cdot \text{larger: } \frac{13}{17}$$

$$51 \quad 2 \frac{1}{3} \text{ 4.04}$$

$$52 \quad \text{bracket: } 1 \cdot = \frac{2}{3}$$

$$53 \quad 3 \frac{1}{3} \times 13 \cdot = 43 \frac{1}{3} \text{ kg}$$

$$54 \quad 2 \text{ 4.07}$$

$$55 \quad \text{enter each fraction} \cdot = \frac{17}{60}$$

$$56 \quad \text{spent: } 88 \div 8 \times 1 = \$11 \cdot \text{left: } 88 - 11 = \$77 \text{ 4.08}$$

4.10 Percentages, fractions and decimals (booklet p. 21)

Examples (model these live)

E1 Write 35% as a fraction.

$$= \frac{35}{100} \cdot = \frac{7}{20}$$

E2 Write $\frac{3}{5}$ as a percentage.

$$= \frac{60}{100} \cdot = 60\%$$

E3 Write 0.08 as a percentage.

$$0.08 \times 100\% \cdot = 8\%$$

Questions (39 on today's skill, 16 from earlier lessons)

$$1 \quad \frac{3}{25}$$

$$2 \quad 1.07$$

$$3 \quad 10\%$$

$$4 \quad \frac{3}{5}$$

$$5 \quad 0.75$$

$$6 \quad 71\%$$

$$7 \quad \frac{3}{4}$$

$$8 \quad 0.05$$

$$9 \quad \frac{14}{3} \text{ 4.01}$$

$$10 \quad 1.43$$

$$11 \quad 14\%$$

$$12 \quad \frac{1}{20}$$

$$13 \quad \text{no 3.11}$$

$$14 \quad = \frac{60}{100} \cdot = 60\%$$

$$15 \quad 0.44$$

$$16 \quad 74\%$$

$$17 \quad 0.3125 \text{ 4.09}$$

$$18 \quad \frac{4}{5}$$

$$19 \quad = \frac{50}{100} \cdot = 50\%$$

$$20 \quad \times 100: 75\% \cdot \frac{75}{100} = \frac{3}{4}$$

$$21 \quad \frac{2}{4} \text{ and } \frac{3}{4} \cdot \frac{1}{2} < \frac{3}{4} \text{ 4.02}$$

$$22 \quad 37.5\%, 40\%, 35\% \cdot 35\%, \frac{3}{8}, 0.4$$

$$23 \quad = \frac{28}{100} \cdot = 28\%$$

$$24 \quad \text{walk 8, ride 4} \cdot \text{bus: } 24 - 8 - 4 = 12 \text{ 4.08}$$

$$25 \quad \times 100: 4\% \cdot \frac{4}{100} = \frac{1}{25}$$

$$26 \quad \times 100: 90\% \cdot \frac{90}{100} = \frac{9}{10}$$

$$27 \quad 40\%, 45\%, 38\% \cdot 38\%, \frac{2}{5}, 0.45$$

- 28 149° 2.04
 29 $= \frac{48}{100} \cdot = 48\%$
 30 $\frac{3}{8}$ 4.03
 31 $= \frac{125}{1000} \cdot = \frac{1}{8}$
 32 $\times 100: 30\% \cdot \frac{30}{100} = \frac{3}{10}$
 33 8 4.07
 34 35%, 40%, 37.5% $\cdot$ 35%, $\frac{3}{8}$, 0.4
 35 $\frac{5}{6}$ 4.01
 36 $= \frac{80}{100} \cdot = 80\%$
 37 $= \frac{75}{1000} \cdot = \frac{3}{40}$
 38 $1 \div 9 \times 100 \cdot \approx 11.1\%$
 39 $2 - (-6) \cdot = 2 + 6 \cdot = 8^\circ\text{C}$ 1.10
 40 7%: the others are all equal.
 41 $\frac{3}{14}$ 4.09
 42 $= \frac{25}{1000} \cdot = \frac{1}{40}$
 43 $1 \div 7 \times 100 \cdot \approx 14.3\%$
 44 $5 + 4 = 9 \cdot \frac{3}{6} + \frac{2}{6} = \frac{5}{6} \cdot = 9$
 45 $\frac{1}{5}$: the others are all equal.
 46 $2 \div 3 \times 100 \cdot \approx 66.7\%$
 47 65%: the others are all equal.
 48 parallel 2.06
 49 $= \frac{875}{1000} \cdot = \frac{7}{8}$
 50 28 4.05
 51 38%, 45%, 40% $\cdot$ 38%, $\frac{2}{5}$, 0.45
 52 $5 \div 6 \times 100 \cdot \approx 83.3\%$
 53 $\frac{7}{18}$ 4.07
 54 $\frac{1}{5}$: the others are all equal.
 55 $25 \div 3 = 8 \text{ r } 1 \cdot = 8\frac{1}{3}$ 4.01

4.11 Percentage of a quantity (booklet p. 23)

Examples (model these live)

- E1 10% of 340
 $340 \div 10 \cdot = 34$
 E2 35% of 80
 $10\% = 8$, so $30\% = 24 \cdot$ $5\% = 4 \cdot$ $35\% = 24 + 4 = 28$
 E3 A \$60 jacket is 25% off. Find the sale price.
 discount: 25% of 60 = \$15 $\cdot$ price: 60 – 15 = \$45

Questions (44 on today's skill, 17 from earlier lessons)

- 1 57
 2 3.5
 3 43.5
 4 68
 5 2.8
 6 8.5
 7 6
 8 40.8
 9 0.2 4.10
 10 12.5
 11 2
 12 88
 13 33.3
 14 71
 15 47
 16 17 3.03
 17 $10\% = 8 \cdot 70\% = 56$
 18 55.3
 19 136
 20 12 4.05
 21 60
 22 $10\% = 20 \cdot 70\% = 140$
 23 $10\% = 4 \cdot 70\% = 28$
 24 50% of 110 $\cdot = 55$ marks
 25 HCF of 28 and 36 is $4 \cdot = \frac{7}{9}$ 4.01
 26 $65 \div 100 \times 460 \cdot = \299
 27 $10\% = 38 \cdot 20\% = 76$
 28 enter each mixed numeral $\cdot = -1\frac{3}{35}$ 4.09
 29 50% of 100 $\cdot = 50$ marks
 30 $35 \div 100 \times 520 \cdot = \182
 31 10% of 190 $\cdot = 19$ marks
 32 $65 \div 100 \times 360 \cdot = \234
 33 4 3.08
 34 $10\% = 34 \cdot 5\% = 17$
 35 $\frac{1}{6}$ 4.02
 36 discount: 40% of 180 = \$72 $\cdot$ price: 180 – 72 = \$108
 37 20% of 70 $\cdot = 14$ marks
 38 $\frac{9}{25} \cdot = \frac{9}{25}$ 4.08
 39 $65 \div 100 \times 240 \cdot = \156
 40 22%, 20%, 25% $\cdot \frac{1}{5}$, 22%, 0.25 4.10
 41 $10\% = 42 \cdot 70\% = 294$
 42 discount: 15% of 160 = \$24 $\cdot$ price: 160 – 24 = \$136
 43 25% of 80 = 20 $\cdot$ 80% of 25 = 20: they are equal
 44 DC + LXX + III $\cdot 600 + 70 + 3 = 673$ 3.06
 45 increase: 25% of 420 = \$105 $\cdot$ new price: \$525
 46 9 4.05
 47 discount: 40% of 220 = \$88 $\cdot$ price: 220 – 88 = \$132
 48 30% of 70 = 21 $\cdot$ 70% of 30 = 21: they are equal
 49 $= \frac{4}{12} + \frac{9}{12} \cdot = 1\frac{1}{12}$ 4.03
 50 increase: 25% of 140 = \$35 $\cdot$ new price: \$175
 51 bracket: $1\frac{5}{18} \cdot = \frac{23}{42}$ 4.09
 52 20% of 45 = 9 $\cdot$ 45% of 20 = 9: they are equal
 53 increase: 5% of 100 = \$5 $\cdot$ new price: \$105
 54 121 3.07
 55 discount: 20% of 110 = \$22 $\cdot$ price: 110 – 22 = \$88
 56 $1\frac{1}{9}$ 4.04
 57 $18 \div 100 \times 600 \cdot = \108
 58 40% of 15 = 6 $\cdot$ 15% of 40 = 6: they are equal
 59 $\frac{1}{3}$, 8 cm 4.08
 60 increase: 25% of 380 = \$95 $\cdot$ new price: \$475
 61 $= \frac{75}{100} \cdot = 75\%$ 4.10

4.12 Expressing as a fraction or percentage (booklet p. 25)

Examples (model these live)

- E1 Write 15 out of 20 as a percentage.
 $\frac{15}{20} = \frac{75}{100} \cdot = 75\%$
 E2 Write 40 cm as a fraction of 2 m.
 $2 \text{ m} = 200 \text{ cm} \cdot \frac{40}{200} = \frac{1}{5}$
 E3 Sam got 18 out of 24. What percentage is that?
 $\frac{18}{24} = \frac{3}{4} \cdot = 75\%$

Questions (39 on today's skill, 17 from earlier lessons)

- 1 30%
 2 $\frac{2}{19}$
 3 40%
 4 $\frac{1}{7}$
 5 40%
 6 $\frac{2}{5}$
 7 32%
 8 $\frac{5}{8}$
 9 24.2 4.11
 10 $\frac{14}{15}$
 11 50%
 12 $\frac{6}{25} = \frac{24}{100} \cdot = 24\%$
 13 right 2.03
 14 $\frac{1}{2}$
 15 20%
 16 $\frac{1}{25} = \frac{4}{100} \cdot = 4\%$
 17 0.49 4.10
 18 $\frac{1}{2}$
 19 $\frac{3}{10} = \frac{30}{100} \cdot = 30\%$
 20 $1 \text{ h} = 60 \text{ min} \cdot \frac{30}{60} = \frac{1}{2}$
 21 HCF of 120 and 156 is 12 $\cdot = \frac{10}{13}$ 4.01
 22 $\frac{19}{20} \times 100\% \cdot = 95\%$
 23 $\frac{20}{25} = \frac{80}{100} \cdot = 80\%$
 24 enter each mixed numeral $\cdot = 2$ 4.09
 25 $2 \text{ m} = 200 \text{ cm} \cdot \frac{10}{200} = \frac{1}{20}$
 26 $1 \text{ h} = 60 \text{ min} \cdot \frac{12}{60} = \frac{1}{5}$
 27 $\frac{28}{40} \times 100\% \cdot = 70\%$
 28 2 3.08
 29 $\frac{5}{10} = \frac{50}{100} \cdot = 50\%$
 30 $\frac{1}{7}$ 4.02
 31 $\frac{36}{60} \times 100\% \cdot = 60\%$
 32 $2 \text{ h} = 120 \text{ min} \cdot \frac{12}{120} = \frac{1}{10}$
 33 138 4.11
 34 $\frac{21}{25} \times 100\% \cdot = 84\%$
 35 70%, 75%, 72% $\cdot$ 0.7, 72%, $\frac{3}{4}$ 4.10
 36 $\frac{19}{20} = \frac{95}{100} \cdot = 95\%$
 37 $\frac{6}{60} \times 100\% \cdot = 10\%$
 38 70% and 72% $\cdot$ better: 36 out of 50

39 list multiples of the largest; test the others · LCM = 12 ^{3.12}

40 $\frac{9}{25} \times 100\% = 36\%$

41 $\frac{3}{7}$ ^{4.09}

42 $\frac{30}{60} \times 100\% = 50\%$

43 65% and 64% · better: 13 out of 20

44 $= \frac{12}{15} - \frac{5}{15} = \frac{7}{15}$ ^{4.03}

45 $\frac{14}{25} \times 100\% = 56\%$

46 $10\% = 26 \cdot 5\% = 13$ ^{4.11}

47 90% and 88% · better: 9 out of 10

48 $\frac{9}{20} \times 100\% = 45\%$

49 they add to 180° ^{2.11}

50 $\frac{48}{60} \times 100\% = 80\%$

51 $2\frac{2}{7}$ ^{4.04}

52 $\frac{21}{40} \times 100\% = 52.5\%$

53 85% and 84% · better: 17 out of 20

54 $\frac{3}{10}$ ^{4.10}

55 $\frac{16}{32} \times 100\% = 50\%$

56 $12 \square / \square 13 \div 5 \square / \square 12 = \cdot$
 $= 2\frac{14}{65}$ ^{4.09}